

Software Requirements Specification (SRS): Civic Lens

1. Project Overview

Project Name: Civic Lens **Purpose:** A highly secure, parallel election result viewing and collation platform designed for political organizations.

2. Core Architectural Principles

- **Zero-Trust Security:** No entity is trusted by default.
 - **Cost-Effective Infrastructure Security:** Prioritize native cloud-provider encryption over computationally expensive application-level encryption for the MVP layer.
 - **Offline-First & PWA Mode:** The system assumes NO internet connection when capturing data.
 - **Modularity:** The codebase (FastAPI & React) will be strictly modularized.
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3. High-Security & Threat Mitigation Requirements

3.1 Network Edge & Transit Security

- **Requirement 3.1.1 (DDoS Mitigation):** Implementation of Cloudflare Enterprise (WAF) and "Under Attack" Bot Management mode.
- **Requirement 3.1.2 (Strict HTTPS):** Enforcement of TLS 1.3 and HSTS for all data in transit padding against Man-In-The-Middle attacks.

3.2 Database Isolation & Encryption at Rest

- **Requirement 3.2.1 (VPC):** PostgreSQL database isolated in a Virtual Private Cloud (VPC) subnet with zero public internet access.
- **Requirement 3.2.2 (Transparent Data Encryption - TDE):** Database volume physically encrypted at rest using native Managed Database Provider settings (e.g., AWS KMS) to ensure hard drive theft is useless.
- **Requirement 3.2.3 (Image Storage Encryption):** S3 Buckets used for storing Form EC8A images will use standard SSE-S3 encryption at rest by default.
- **Requirement 3.2.4:** Result records follow **WORM** principles. Alterations trigger immutable Audit Logs.
- **Requirement 3.2.5:** Mandatory Multi-Factor Authentication (MFA) for dashboard access.

3.3 Advanced Agent Security Protocols

- **Requirement 3.3.1 (Duress Password/Panic Mode):** Agents can enter a secondary PIN if coerced physically. The backend discretely flags the payload as compromised.
- **Requirement 3.3.2 (Device Fingerprinting):** Agent accounts are strictly bound to their hardware MAC/IMEI equivalent.

4. Deep-Offline & Poor Connectivity Requirements

4.1 Client-Side Payload Optimization

- **Requirement 4.1.1 (Image Compression):** Shrink captured images down to < 300KB via WASM/JS libraries before network transmission.

4.2 Progressive Web App (PWA) Offline Engine

- **Requirement 4.2.1 (IndexedDB Storage):** Form data streams locally into IndexedDB.
- **Requirement 4.2.2 (Background Sync Queue):** Network requests that timeout enter a Background Sync Queue utilizing Exponential Backoff.

4.3 Fallback Transmission Protocols

- **Requirement 4.3.1 (SMS Fallback):** If TCP/IP fails, the app encodes the vote data into a dense string and triggers the native SMS app.
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5. System Features & Workflows

5.1 Field Agent Module (Frontend)

- **Feature 5.1.1:** Authentication with short-lived JWT tokens.
- **Feature 5.1.2:** Soft Geofencing. GPS coordinate appending without blocking uploads.

5.2 Situation Room Verification Module (Admin Dashboard)

- **Feature 5.2.1:** VPN / IP-Restricted access.
- **Feature 5.2.2 (AI & OCR Cross-Checking):** An OCR engine reads handwritten numbers from the Form EC8A image and locks discrepant data into a "High-Risk Queue".

5.3 Executive Collation Dashboard

- **Feature 5.3.1:** High-performance caching layer (Redis) executing rolling sums.
- **Feature 5.3.2:** Interactive choropleth heat-maps utilizing real-time SSE updates.